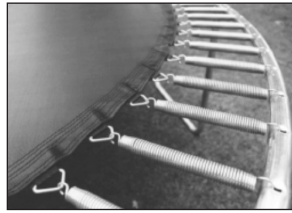
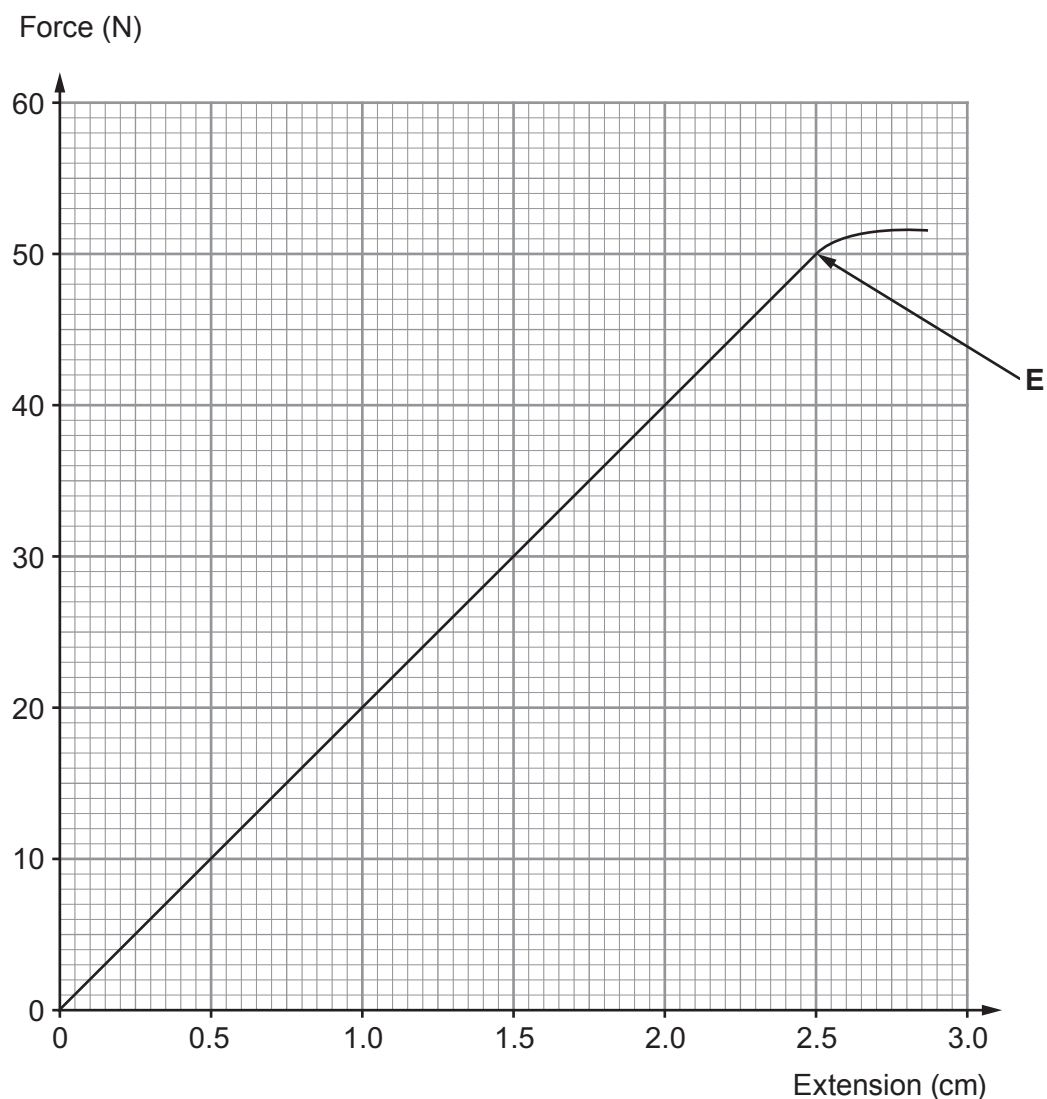


3. An engineering firm in South Wales makes springs for trampolines.



A trampoline is an elastic disc that is connected to a frame by many springs connected in parallel. As you land on a trampoline the springs stretch. The springs work to return to their normal length. The springs pull back against your weight as you land. The heavier the person on the trampoline, the longer the springs extend. Springs are tested to see if they return to their normal length after being stretched. The spring will be permanently stretched if it extends beyond the elastic limit (point **E**) where Hooke's Law is no longer obeyed.

The graph shows how far a particular spring extends when forces are applied.



- (a) (i) Use the graph to find the largest force at which Hooke's Law is still obeyed. [1]

Force = N

- (ii) Each spring in one trampoline will experience a maximum force of 58 N.
Explain whether this particular spring is suitable for use in the trampoline. [2]

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- (b) (i) Calculate the spring constant for this spring up to point **E** using the equation: [3]

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

Spring constant = N/cm

- (ii) **Add a line** to the graph for a spring with a smaller spring constant. [2]

